VLSI.AI

AI-Powered Interview Platform

Product Documentation

Version 1.0

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Confidential

1. Executive Summary

VLSI.AI is a fully automated, voice-based AI interview platform purpose-built for the VLSI semiconductor industry. It conducts real-time technical interviews with candidates using an AI interviewer persona, evaluates responses against calibrated rubrics, and generates detailed scoring reports -- eliminating the need for human interviewers in the screening round.

Key value propositions:

- Replaces 30-45 minute human screening interviews with a consistent AI interviewer
- Covers 3 VLSI domains across 2 experience levels (6 interview profiles)
- Real-time voice interaction with sub-second latency (STT + TTS pipeline)
- Multi-layered anti-cheat: face verification, speaker verification, gaze tracking, AI-answer detection, tab-switch monitoring
- LMS-ready with API-based session launch and score callback
- Full observability: per-call cost tracking, latency metrics, and admin review tools

2. Product Overview

What It Does

The platform conducts a live, voice-based technical interview where:

- A candidate uploads their resume and enters the interview room
- An AI interviewer (with a distinct persona and voice) greets them and begins asking domain-specific questions
- The AI adapts questions based on the candidate's resume, prior answers, and experience level
- After 12-18 turns, the AI ends the interview and generates a structured evaluation
- Admins and hiring managers review scores, transcripts, and anti-cheat signals via the admin panel

Who Uses It

User	Interface	Purpose
Candidates	Web browser (desktop)	Take the voice interview
Hiring Managers / Admins	Admin panel	Review sessions, scores, transcripts, a
LMS Systems	REST API	Launch interviews, receive score callba

3. System Architecture

High-Level Architecture

```
                    +--------------------+
                    |   Candidate Browser |
                    |  (voice_agent_ui)   |
                    +--------+-----------+
                             |
                     HTTPS / WSS
                             |
                    +--------v-----------+
                    |   FastAPI Server    |
                    |   (uvicorn, multi-  |
                    |    worker capable)  |
                    +--------+-----------+
                             |
      +----------+-----------+-----------+----------+
      |          |           |           |          |
+-----v----+ +--v---+ +----v----+ +----v----+ +---v----+
|PostgreSQL | |Redis | |LLM APIs | |STT APIs | |TTS APIs|
| (primary  | |(cache| |         | |         | |        |
|  storage) | | +cfg)| |         | |         | |        |
+----------+ +------+ +---------+ +---------+ +--------+
                             |
                      +------v-------+
                      | AWS Services |
                      | Rekognition  |
                      | Bedrock      |
                      | Textract     |
                      | SSM / SM     |
                      +--------------+
```

Tech Stack

Layer	Technology
Backend	Python 3.11+, FastAPI, Uvicorn (ASGI)
Database	PostgreSQL (primary), Redis (cache + cross-worker config)
Frontend	Vanilla HTML/JS, Tailwind CSS, MediaPipe, TensorFlow.js
LLM	OpenAI GPT-4o-mini, AWS Bedrock (Claude, Llama, Nova), xAI Grok, Cerebras
STT	OpenAI Whisper, Deepgram Nova, Inworld STT
TTS	Deepgram Aura, Kugel, Inworld, OpenAI TTS
Cloud	AWS EC2, Rekognition, Textract, SSM Parameter Store, Secrets Manager
Auth	JWT (HS256), cookie + bearer token

Data Flow

```
Resume Upload -> Parse (Cerebras/LLM) -> Create Session -> Store in PostgreSQL
                                                              v
Interview Start -> LLM Greeting -> TTS Synthesis -> Audio to Browser
                                                              v
Candidate Speaks -> Browser VAD (1s pause) -> Audio Chunk -> STT API
                       v                                      v
              (30s auto-chunk mid-speech)              Transcript
                                                              v
                                              Submit Answer -> LLM generates
                                              next question -> TTS -> Browser
                                                              v
                                              (Background: AI detection,
                                               speaker verify, expected
                                               points generation)
                                                              v
                                              Interview Ends -> Evaluate
                                              via LLM -> Store scores ->
                                              LMS callback (if applicable)
```

4. Features

4.1 Voice-Based Interview

- Real-time voice interaction using browser microphone
- Browser-side Voice Activity Detection (VAD) with configurable silence thresholds
- 1-second pause: commits audio chunk to STT in background (non-blocking)
- 2-second total silence: finalizes the turn and submits answer
- 30-second auto-chunking for long answers (sends mid-speech, zero-gap restart)
- Accumulated transcript from multiple chunks combined before submission
- Streaming LLM response with real-time TTS synthesis (sentence-level chunking)

4.2 Adaptive AI Interviewer

- 6 distinct interviewer personas across domains and levels
- Each persona has a defined name, personality, speech style, and word limits
- Questions adapt to candidate's resume content (only asks about listed skills/projects)
- Three question types enforced: Project (40%), Scenario/Debug (40%), Concept (20%)
- Follow-up probing when expected answer points are missed
- Deliberate incorrect statements to test candidate's ability to push back
- Dynamic pacing: end at 12-15 turns, extend to 18 for strong candidates, cut at 10 for weak
- Returning candidate awareness: fresh questions for repeat interviews

4.3 Resume Parsing

- Supports PDF, DOCX, and TXT formats
- PDF extraction via pdfplumber with AWS Textract fallback for scanned documents
- Fast parsing via Cerebras (Llama 3.1-8b) with fallback to primary LLM
- Extracts: name, email, experience years, domain, skills, tools, projects, education

- 3-attempt retry with validation for robust extraction

4.4 Evaluation & Scoring

- Post-interview evaluation via LLM against calibrated rubrics
- Separate rubrics for junior (1-5 years) and senior (5+ years) candidates
- Per-question scoring with quality, accuracy, and quadrant classification
- Expected points comparison (generated in background during interview)
- Follow-up questions grouped with parent for combined scoring
- Output: overall score, grade, per-topic breakdown, strengths, weaknesses
- Minimum 8 answered questions required for evaluation

4.5 Face Verification (AWS Rekognition)

- Reference face registration during lobby (webcam capture)
- Glasses detection with user notification
- Periodic face comparison during interview against stored reference
- Configurable similarity threshold (default: 90%)
- Stored per-email for returning candidates

4.6 Speaker Verification (Resemblyzer)

- Voice embedding computed from first answer (256-dimensional vector)
- Subsequent answers compared against reference embedding
- Two-strike system: first mismatch is a warning, second triggers session termination
- Cosine similarity threshold: 0.75
- Minimum 3 seconds of audio required for reliable embedding

4.7 Gaze Tracking

- Client-side: MediaPipe Face Mesh extracts 90 facial landmark features
- Server-side: ExtraTrees classifier (sklearn) classifies gaze as left/right/straight
- Logs gaze deviations as anti-cheat signals

4.8 AI Answer Detection

- Dual detection: Sapling AI API + LLM cross-verification
- Runs in background thread per answer (non-blocking)
- Results stored in session for admin review

4.9 Browser Anti-Cheat

- Tab switch / window blur detection
- Copy/paste monitoring
- DevTools open detection
- Right-click context menu blocking
- Object detection via TensorFlow.js COCO-SSD (detects phones, books, second screens)
- Anti-cheat canary element (invisible text to detect AI page scrapers)

- All events logged with timestamps for admin review

4.10 Shared Review Links

- Admins can generate time-limited read-only links (default: 72 hours)
- Separate JWT with type: share -- no admin access
- Accessible without login at /review/{token}

5. Interview Flow

5.1 Candidate Journey

```
1. LOBBY (index.html)
   +-- Enter name, email, years of experience
   +-- Select domain (Analog Layout / Physical Design / Design Verification)
   +-- Upload resume (PDF/DOCX/TXT)
   +-- Resume parsed and validated
   +-- Face registration (webcam capture, if enabled)
   +-- Click "Start Interview"

2. INTERVIEW (voice_agent_ui.html)
   +-- AI generates personalized greeting (time-of-day aware)
   +-- WARM_OPENING phase (turns 1-3): Light questions to build comfort
   +-- DISCOVERY phase (turns 4-7): Explore breadth of knowledge
   +-- ADAPTIVE_DEPTH phase (turns 8+): Deep dive based on responses
   +-- Each turn:
   |   +-- Candidate speaks -> audio captured -> STT -> transcript
   |   +-- Background: speaker verification, AI detection
   |   +-- LLM generates next question (streaming)
   |   +-- TTS synthesizes response (streaming, sentence-level)
   |   +-- Background: expected points generated for scoring
   +-- Interview ends (LLM decision, time limit, or turn limit)

3. EVALUATION (automatic)
   +-- Full transcript built with expected points
   +-- LLM evaluates against level-calibrated rubric
   +-- Per-question scores + overall score + grade computed
   +-- Results persisted to PostgreSQL
   +-- LMS callback triggered (if LMS-launched session)

4. REPORT
   +-- Candidate can view their evaluation summary
```

5.2 Interview Ending Conditions

Condition	Trigger	Source
LLM decision	AI determines interview is complete (12	LLM via [END_INTERVIEW] tag
Turn limit	25 turns reached	Backend hard limit
Time limit	1 hour elapsed	Backend hard limit
Stale session	No activity for 1 hour	Background sweeper
Abuse detected	Candidate uses abusive language	LLM via [ABUSIVE] tag
Speaker mismatch	Voice doesn't match reference (2 strike	Speaker verification
Manual end	Candidate clicks "End Interview"	Frontend

6. AI & LLM Integration

6.1 LLM Providers

Provider	Models	Use Case
OpenAI	GPT-4o-mini (default)	Question generation, evaluation
AWS Bedrock	Claude Haiku/Sonnet/Opus, Llama 4, Amaz	Alternative question generation and eva
xAI	Grok 4.1 (fast reasoning / non-reasonin	Alternative LLM provider
Cerebras	Llama 3.1-8b	Fast resume parsing (low latency)

All LLM providers are hot-swappable via the admin panel without restart.

6.2 STT Providers

Provider	Models	Features
OpenAI	gpt-4o-mini-transcribe, whisper-1	Default provider
Deepgram	nova-3, nova-2	VLSI keyword boosting (50 domain terms,
Inworld	inworld-stt-1	Alternative provider

All providers pass through a centralized hallucination filter that catches common STT artifacts (e.g., "Thank you for watching", "Please subscribe").

6.3 TTS Providers

Provider	Voices	Features
Deepgram	aura-asteria-en (default)	Primary, low latency
Kugel	kugel-2.5	Raw PCM -> WAV conversion
Inworld	Configurable	Alternative
OpenAI	tts-1	Fallback

6.4 Domain-Specific STT Prompts

Each domain has a custom STT prompt file (stt_prompts/) containing domain-specific vocabulary to improve transcription accuracy for technical VLSI terms.

7. Anti-Cheat & Proctoring System

Feature Matrix

Feature	Client-Side	Server-Side	Toggleable
Face verification	Webcam capture	AWS Rekognition comparison	Yes
Speaker verification	Audio capture	Resemblyzer embedding comp	Yes
Gaze tracking	MediaPipe Face Mesh (90 fe	ExtraTrees classifier	Yes
AI answer detection	--	Sapling API + LLM cross-ch	Yes
Tab switch detection	Visibility API	Event logging	Yes
Copy/paste detection	Clipboard events	Event logging	Yes
Object detection	TensorFlow.js COCO-SSD	--	Yes
DevTools detection	Window size heuristics	--	Yes
Content canary	Hidden DOM element	--	Always on

All anti-cheat features are individually toggleable via the admin panel.

8. Admin Panel

Capabilities

Section	Features
Dashboard	Live session count, total interviews, aggregate stats
Sessions	List all sessions with scores, grades, anti-cheat flags; search/filter
Session Detail	Full transcript, per-question scores, evaluation breakdown, anti-cheat ti
LLM Config	Switch question-generation and evaluation models on the fly
Prompt Editor	View and edit evaluation prompts per level; reset to defaults
STT Config	Switch STT provider and model
TTS Config	Switch TTS provider and voice; test TTS with custom text
STT Playground	Upload audio and test transcription with any provider/model
Prompt Playground	Test any prompt against any model with adjustable temperature
Anti-Cheat Config	Toggle individual anti-cheat features
Voice Verification	Enable/disable speaker verification globally
Observability	Aggregated LLM/STT/TTS costs, latency stats, per-call logs
Share Links	Generate time-limited read-only session review links
Re-run Evaluation	Re-evaluate any session with current model/prompts

9. LMS Integration

Launch Flow

```
LMS System                          VLSI.AI Platform
    |                                      |
    |  POST /api/lms/launch                |
    |  X-API-Key: <lms_api_key>            |
    |  {name, email, domain,               |
    |   resume_url/resume_base64,          |
    |   voice, user_face}                  |
    |------------------------------------->|
    |                                      |  Parse resume
    |                                      |  Create session
    |                                      |  Store face reference
    |  {session_id, interview_url}         |
    |<-------------------------------------|
    |                                      |
    |  Candidate opens interview_url       |
    |  (signed JWT, 30-min expiry)         |
    |------------------------------------->|
    |                                      |
    |        ... interview happens ...     |
    |                                      |
    |  POST <lms_callback_url>             |
    |  {session_id, candidate_name,        |
    |   candidate_email, domain, level,    |
    |   overall_score, grade, status,      |
    |   per_question scores, strengths,    |
    |   weaknesses, topics}              |
    |<-------------------------------------|
```

LMS Launch Parameters

Parameter	Type	Required	Description
name	string	Yes	Candidate full name
email	string	Yes	Candidate email
domain	string	Yes	Interview domain
resume_url or resume_base6	string	Yes (one of)	Resume file
voice	string	No	TTS voice override
callback_url	string	No	Score delivery URL
user_face	string	No	Base64 face reference imag
level	string	No	Override auto-detected lev
years_experience	number	No	Override years

10. Security

Authentication & Authorization

Mechanism	Scope	Details
JWT (HS256)	Admin panel	8-hour expiry, bearer + cookie support
JWT (HS256)	LMS launch	30-minute expiry, single-use signed URL
JWT (HS256)	Share links	72-hour expiry, read-only access
API Key	LMS API	X-API-Key header validation
Admin credentials	Login	Username/password from secrets store

Secrets Management

Secrets are never hardcoded. The secrets_proxy module abstracts secret retrieval with three backends:

Backend	Environment	Source
env	Development	.env file / environment variables
ssm	Production	AWS SSM Parameter Store (SecureString)
secretsmanager	Production	AWS Secrets Manager

Backend selection is automatic via the SECRETS_BACKEND environment variable.

Data Protection

- Session data stored as JSONB in PostgreSQL (encrypted at rest via AWS)
- Face reference images stored as BYTEA in PostgreSQL
- Redis cache has 2-hour TTL; invalidated on sensitive updates
- No candidate PII in application logs
- Share links are time-limited and read-only

11. Infrastructure & Deployment

Current Deployment

Component	Details
Server	AWS EC2 instance
Application	Uvicorn (multi-worker capable)
Database	PostgreSQL (connection pool: 1-10)
Cache	Redis
Deployment method	Git push -> git pull on EC2 -> restart

Database Schema

12 tables managing the complete interview lifecycle:

Table	Purpose
candidates	Candidate profiles (name, domain, level, skills, tools)
sessions	Completed session metadata and scores
turns	Per-turn question, answer, duration, word count
evaluations	Per-turn scores, quality, accuracy, expected/missing points
behavioral_signals	Per-turn filler rate, pause times, behavioral flags
reports	Aggregate session scores by category
expert_reviews	Human reviewer overlays (for calibration)
llm_calls	Per-call LLM/STT/TTS cost and latency logging
active_sessions	Live session state (JSONB blob)
app_config	Runtime configuration (shared across workers)
candidate_history	Per-email session history for returning candidates
face_references	Stored face images for verification

Background Processes

Process	Interval	Purpose
Evaluation sweeper	Every 5 minutes	Catch sessions that missed foreground e
Stale session sweeper	Continuous	End sessions inactive > 1 hour
Runtime config sync	On startup	Sync admin config across workers

12. API Reference

Candidate-Facing Endpoints

Method	Endpoint	Description
GET	/	Interview lobby page
GET	/interview	Interview room page
GET	/health	Health check
GET	/api/lobby-config	Get lobby feature flags
GET	/api/domains	List available interview domains
GET	/api/tts-status	Check if TTS is enabled
POST	/api/parse-resume	Upload and parse resume
POST	/api/create-session	Create new interview session
POST	/api/start-interview	Start interview, get greeting
POST	/api/transcribe	Transcribe audio chunk
POST	/api/submit-answer	Submit answer (non-streaming)
POST	/api/stream-answer	Submit answer (streaming SSE)
POST	/api/end-session	End session manually
GET	/api/get-session	Get session metadata
POST	/api/generate-report	Get evaluation report
POST	/api/anticheat-event	Log anti-cheat event
GET	/api/anticheat-settings	Get anti-cheat feature states
POST	/api/anticheat/gaze	Classify gaze direction
GET	/api/face/check	Check face reference exists
POST	/api/face/register	Register face reference
POST	/api/face/detect-glasses	Detect glasses
POST	/api/face/compare	Compare face against reference
WS	/ws/audio	WebSocket audio stream

Admin Endpoints (JWT Required)

Method	Endpoint	Description
GET	/admin	Admin panel page
POST	/api/auth/login	Admin login
POST	/api/auth/logout	Admin logout
GET	/api/auth/me	Validate current session
GET	/api/admin/sessions	List all sessions
GET	/api/admin/session/{sid}	Session detail
POST	/api/admin/rerun-eval/{sid}	Re-run evaluation
GET/POST	/api/admin/llm-config	Get/set LLM models
GET/POST	/api/admin/llm-prompts	Get/set evaluation prompts
GET	/api/admin/qgen-prompt	Get interviewer prompt
GET	/api/admin/interview-prompt	Get raw prompt file
GET/POST	/api/admin/stt-config	Get/set STT config
POST	/api/admin/stt-test	STT playground
GET/POST	/api/admin/voice-verification	Get/toggle speaker verification
POST	/api/admin/set-interview-voice	Set TTS voice
POST	/api/admin/test-tts	TTS playground
POST	/api/admin/prompt-playground	Prompt playground
GET/POST	/api/admin/anticheat-config	Get/toggle anti-cheat features
POST	/api/toggle-tts	Toggle TTS globally
POST	/api/admin/share-link	Generate share link
GET	/api/shared/session/{token}	Get shared session
GET	/api/observability/summary	Cost/latency summary
GET	/api/observability/logs	Raw observability logs

LMS Endpoints

Method	Endpoint	Description
POST	/api/lms/launch	Pre-create session from LMS

13. Cost & Observability

Per-Call Tracking

Every LLM, STT, and TTS call is logged with:

- Model/provider used
- Input/output token counts (LLM) or audio duration (STT) or character count (TTS)
- Latency in milliseconds

- Computed USD cost
- Success/error status

Cost Estimation Models

Service	Pricing Basis
LLM	Per-token (input + output), model-specific rates
STT	Per-minute of audio, model-specific rates
TTS	Per-character, provider-specific rates

Admin Observability Views

- Summary: Aggregate cost by step (question generation, evaluation, STT, TTS), average latency, total calls
- Logs: Raw per-call breakdown filterable by session, step, model, and time range

14. Supported Domains

Interview Domains

Domain	Junior Prompt	Senior Prompt	STT Prompt
Analog Layout	Yes	Yes	Yes
Physical Design	Yes	Yes	Yes
Design Verification	Yes	Yes	Yes

Experience Levels

Level	Years	Calibration
Experienced Junior	1-3 years	Expects hands-on tool usage, real debug
Experienced Senior	3+ years	Expects tradeoff reasoning, architectur

Each level has distinct interviewer personas, question depth expectations, evaluation rubrics, and scoring calibration.

File Structure

```
simple_interview/
+-- main.py                     # Backend: FastAPI app, all endpoints (4,100 lines)
+-- config.py                   # Configuration, AI client init, domain constants
+-- database.py                 # PostgreSQL operations, connection pool
+-- redis_cache.py              # Redis session cache, cross-worker config
+-- secrets_proxy.py            # Secrets abstraction (env / SSM / Secrets Manager)
+-- schema.sql                  # PostgreSQL schema (12 tables)
+-- requirements.txt            # Python dependencies
+-- loadtest.py                 # Load testing utility
+-- .env                        # Local environment variables
|
+-- prompts/                    # Live interviewer system prompts
|   +-- experienced_junior_analog_layout.md
|   +-- experienced_junior_design_verification.md
|   +-- experienced_junior_physical_design.md
|   +-- experienced_senior_analog_layout.md
|   +-- experienced_senior_design_verification.md
|   +-- experienced_senior_physical_design.md
|
+-- eval_prompts/               # Post-interview evaluation rubrics
|   +-- experienced_junior.md
|   +-- experienced_senior.md
|
+-- stt_prompts/                # Domain-specific STT vocabulary
|   +-- analog_layout.txt
|   +-- design_verification.txt
|   +-- physical_design.txt
|
+-- models/                     # ML models
|   +-- gaze_ensemble_model.pkl    # Gaze direction classifier
|   +-- spkrec-ecapa/              # Speaker recognition model
|
+-- templates/                  # Frontend HTML
    +-- index.html              # Candidate lobby (1,364 lines)
    +-- voice_agent_ui.html     # Interview room (3,022 lines)
    +-- admin.html              # Admin panel (3,396 lines)
    +-- shared_review.html      # Read-only review page
    +-- lms_test.html           # LMS integration test page
    +-- speaker_test.html       # Speaker verification test page
```